

# XRPL Validator — Daily Engineering Report • 2026-09-14

---

Compiled retrospectively on 2026-09-23 from the git history. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the XRP Ledger itself (amendments) and XRPL research.

## Summary

---

- 24 commits (xrpl-validator 24)
- 9 engine findings fixed (F278–F289)
- 1 test / vector commit

## Engine findings fixed

---

- **fix(xrpl-ledger): F278 — the per-iteration fold rests an offer only when its level list accounts for every fill** ( 678f5d5 , xrpl-validator) Finding 274 folded the rested amount over `saved_level_ins` (rippled's per-iteration `savedIns`). A fill path that spends the taker's whole remaining budget records no level entry — the in side is closed by construction (Finding 30's drain law) and level folding stops — so on an exhausted walk the level list is short of the fills and the fold rests what was never left over. The tail now uses the level fold only when ...
- **feat(ledger): port rippled's Number at the 16-digit scale; route IOU add/sub through it (F280b, F281)** ( e7beae9 , xrpl-validator) Number's mantissa width is amendment-gated (Rules.cpp `setCurrentTransactionRules`): the 19-digit Large scale switches on with `SingleAssetVault` or `LendingProtocol`, neither enabled on mainnet, so transaction processing runs the 16-digit Small scale with the cusp-rounding fix disabled. The arith shim was answering at the process default (Large330) — every "19-digit double rounding" row in the first fuzz was that. ...
- **fix(xrpl-ledger): F282 — the bridged crossing's direct strand steps its book by the want, reaping the dead offers it reaches** ( 28c8adc , xrpl-validator) #106983394 34561BC0DC71: rsPrWzpYp5 buys 9862 USDC.rcEGREd8 with RLUSD from a 316.5 RLUSD line, autobridged. rippled's direct strand sizes its `BookStep` in the rev pass against the want (finding 114): rUgWjzwo's head offer is consumed whole in the sandbox (399.04 USDC), the stream steps to rDeXHakZ's ECB6743009C2 behind it, funded by a zero line — "Removing unfunded offer", a `permRmOffer` that outlives the pass — and ...
- **fix(xrpl-ledger): F283 — an empty book directory page is no level: the tip, the pool's yardstick and the walk skip it** ( 8468b9c , xrpl-validator) #106983394 5B4E06B079D4 (rapido5rxP, RLUSD → USD → CNY → XRP, partial with limit quality): the CNY.rKiCet8/XRP book carries an EMPTY root page at 4D0ADF550834D180 — better than every resting offer and than the pool's spot. Our level list took it as the tip: the pool's turn measured against it ("CLOB better") and never produced the slice, and the walk went on to consume rN5nsAc5's and rwvLbHQ's offers at 4F21... ...
- **fix(xrpl-ledger): F285/F287 — after a pool-served pass the stream steps the book only while the flow's SendMax budget lasts; F286 — credentials::valid in Payment preclaim** ( 2b4b9ac , xrpl-validator) Finding 285 (#106983955 D4DD62F77155): rDsxxqzXU pays itself 9999999999999999 XRP for USD.rKiCet8 from a 29.99 USD line with `SendMax` 1e15. The pool's slice at the tip's quality takes everything the taker holds; rippled's

flow then runs another iteration (remainingIn — the SendMax budget — is still positive), whose rev pass steps the book past the slice and permRmOffers rGSooBxyk's EXPIRED 657585B0C18C on the next ...

- **fix(xrpl-ledger): F288 — a bridge leg's admission bound reads the raw BookTip, expired or not** ( 7fe7b60 , xrpl-validator) #106990975 36B71A8A3816: rMsXVzCug7's tfPassive 0.02447 BTC → 1907.06654 RLUSD, bridged through the BTC/XRP and XRP/RLUSD pools. rippled admits a strand on BookStep::tip, whose CLOB half is BookTip::step — the first directory entry as it stands, with no expiry or funding test (the "Removing expired offer" step belongs to the OfferStream and only runs once the strand is flowed). Leg B's raw tip E21DCFAF had ...
- **fix(xrpl-ledger): F289 — a payment's in-driven pass steps the book by its carry, and a fill trimmed on the in side ends the walk** ( cc5a30e , xrpl-validator) #106991212 0B722FF6EA7F: r9tcGwSyYP pays itself 0.071 RLUSD for 50000 drops, XRP → XUSD → USDC.axl → RLUSD, the first two hops pool-served. The last hop's forward pass carries 0.0710127 USDC.axl onto rpkHXWZu's 97.3 head offer. rippled's fwdImp consumes an offer whole only while stpAmt.in <= remainingIn and stops the moment an offer had to be trimmed to the remaining input ( limitStepIn , processMore = false): a ...

## Engineering changes

---

- **feat(ffi,ledger): track 1 — arithmetic oracle exports, differential fuzz harness, two exact primitives** ( 639e778 , xrpl-validator) The shim gains ffi/xrpl\_arith.cpp: one C export per STAmount operation (mulRound, mulRoundStrict, divRound, divRoundStrict, multiply, divide, add, sub, canonicalize), getRate, and Number ops under the four rounding modes — pure functions of their arguments, no ledger. The shim build now matches rippled 3.3.0 (C++23 for std::expected; mpt-crypto from conan for Protocol.h's ).
- **feat(ledger): AMM drops conversion through Number::to\_drops; fuzz STAmount multiply/divide and Number → drops** ( 94e6904 , xrpl-validator) The shim exports Number's explicit int64 conversion (XRPAmount{Number}, operator rep() : fraction digits through the guard, the mode decides the drop) and the fuzz puts it, STAmount multiply (a Number product at nearest) and STAmount divide (the legacy muldiv(num, 1e17, den) + 5 quotient) through libxrpl and the engine: 31 rows × 20,000 cases, zero mismatches.
- **feat(ffi,node): Batch leg A — apply inner transactions inside the outer's shim apply; skip inner ledger entries in the FFI feed** ( 06b21f8 , xrpl-validator)
- **feat(node): count inner Batch entries as covered — stats field, metrics line, parity\_probe completeness** ( 964a862 , xrpl-validator) Eight of the eleven devnet Batch ledgers 5309212–5309670 probe CLEAN through the outer with every inner folded; the three DIVERGENT ones fail on VaultDeposit / LoanPay hydration (SingleAssetVault / LendingProtocol, devnet-only), not on Batch.
- **feat(node): vote yes on BatchV1\_1** ( 58d8f65 , xrpl-validator)
- **feat(node): vote yes on fixCleanup3\_3\_0** ( b26b17b , xrpl-validator)
- **feat(ledger): track 2 slice 1 — flow module: amounts, QualityFunction, PaymentSandbox, the Step trait** ( 1feacf6 , xrpl-validator) The structural port of rippled 3.3.0's path engine begins, behind a module nothing dispatches to yet: - flow::amounts — IOUAmount (a signed 16-digit Number with STAmount's exponent clamp; add/sub are tx::number at ToNearest), XRPAmount, EitherAmount. - flow::quality\_function — QualityFunction.h/.cpp: the CLOB-like constant function, the AMM line through the pool balances at the fee, combine, and outFromAvgQ under ...

## Tests and campaigns

---

- **test(xrpl-node): drop a stray brace appended after the F282 vector (gate131 SUITES build error)**  
( c089330 , xrpl-validator)

## Docs and operations

---

- **docs: Batch (BatchV1\_1) support design — the FFI leg finding, rippled semantics, two-leg plan** ( a33e1d6 , xrpl-validator)
- **docs: Batch spec — leg A verification status** ( 6a7f0bd , xrpl-validator)
- **flow: slice 3 — AMMContext, AMMLiquidity::getOffer, AMMOffer, wired into BookStep** ( a2d060a , xrpl-validator)
- **flow: slice 2 (part 2) — BookStep: forEachOffer, revImp, fwdImp, limitStepIn/Out, consumeOffer**  
( 01aa737 , xrpl-validator)
- **flow: slice 2 (part 1) — BookTip and FlowOfferStream, TOffer, Quality::ceil\_in/out** ( a904cff , xrpl-validator)
- **flow: slice 1b — STAmount rounding port, pinned by fuzz; slice 2 opens with view.rs** ( c195d83 , xrpl-validator)
- **flow: slice 4 (part 1) — DirectStepI, XRPEndpointStep, rippleCredit/rippleSend/accountSend over the sandbox, XRP deferred credits** ( 9743159 , xrpl-validator)
- **flow: slice 4 (part 2) — StrandFlow (single- and multi-strand flow, ActiveStrands, limitOut) and PaySteps (toStrand / toStrands)** ( e0fa870 , xrpl-validator)
- **flow: slice 4 (part 3) — Flow.cpp's flow(), rippleCalculate, and the Payment dispatch behind XRPL\_FLOW\_ENGINE=port** ( 3f5a90e , xrpl-validator)

## Amendment watch

---

*Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.*

No amendment held a majority.

Amendments enabled on mainnet: 94.

## XRPL research

---

### Automated research — candidate amendment ideas

*Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: [iceland.halcyon-names.io](https://iceland.halcyon-names.io)). They are exploration leads, not proposals.*

- **Time-Fair Refilling Iceberg Offers (dex)** Add a tfIceberg flag to OfferCreate plus two new Offer SFields — sfDisplayQuantity (the visible clip, quantized) and sfConcealedQuantity (the hidden r

- **Atomic AMM Flash Loans via Batch** (lending) Add an AMMFlashBorrow inner-transaction type that draws sfAmount of a pool asset directly from an AMM's pseudo-account without burning LP tokens, perm
- **Attestation-Scored Vault Collateral Factors** (lending) Add an AttestationSet transaction and ltATTESTATION ledger object (owner-reserved, keylet::attestation(attestor, subject)) carrying sfScore (uint16) a

## Appendix — commits

| Time  | Repo           | Commit  | Subject                                                                                                                                                                    |
|-------|----------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10:41 | xrpl-validator | 678f5d5 | fix(xrpl-ledger): F278 — the per-iteration fold rests an offer only when its level list accounts for every fill                                                            |
| 11:03 | xrpl-validator | 639e778 | feat(ffl,ledger): track 1 — arithmetic oracle exports, differential fuzz harness, two exact primitives                                                                     |
| 11:40 | xrpl-validator | e7beae9 | feat(ledger): port rippled's Number at the 16-digit scale; route IOU add/sub through it (F280b, F281)                                                                      |
| 11:56 | xrpl-validator | 94e6904 | feat(ledger): AMM drops conversion through Number::to_drops; fuzz STAmount multiply/divide and Number → drops                                                              |
| 12:12 | xrpl-validator | 06b21f8 | feat(ffl,node): Batch leg A — apply inner transactions inside the outer's shim apply; skip inner ledger entries in the FFI feed                                            |
| 12:12 | xrpl-validator | a33e1d6 | docs: Batch (BatchV1_1) support design — the FFI leg finding, rippled semantics, two-leg plan                                                                              |
| 12:17 | xrpl-validator | 964a862 | feat(node): count inner Batch entries as covered — stats field, metrics line, parity_probe completeness                                                                    |
| 12:20 | xrpl-validator | 6a7f0bd | docs: Batch spec — leg A verification status                                                                                                                               |
| 12:30 | xrpl-validator | 58d8f65 | feat(node): vote yes on BatchV1_1                                                                                                                                          |
| 12:32 | xrpl-validator | b26b17b | feat(node): vote yes on fixCleanup3_3_0                                                                                                                                    |
| 12:48 | xrpl-validator | 28c8adc | fix(xrpl-ledger): F282 — the bridged crossing's direct strand steps its book by the want, reaping the dead offers it reaches                                               |
| 13:00 | xrpl-validator | 8468b9c | fix(xrpl-ledger): F283 — an empty book directory page is no level: the tip, the pool's yardstick and the walk skip it                                                      |
| 20:26 | xrpl-validator | 2b4b9ac | fix(xrpl-ledger): F285/F287 — after a pool-served pass the stream steps the book only while the flow's SendMax budget lasts; F286 — credentials::valid in Payment preclaim |
| 21:15 | xrpl-validator | 7fe7b60 | fix(xrpl-ledger): F288 — a bridge leg's admission bound reads the raw BookTip, expired or not                                                                              |
| 21:32 | xrpl-validator | cc5a30e | fix(xrpl-ledger): F289 — a payment's in-driven pass steps the book by its carry, and a fill trimmed on the in side ends the walk                                           |
| 22:05 | xrpl-validator | c089330 | test(xrpl-node): drop a stray brace appended after the F282 vector (gate131 SUITES build error)                                                                            |
| 22:05 | xrpl-validator | a2d060a | flow: slice 3 — AMMContext, AMMLiquidity::getOffer, AMMOffer, wired into BookStep                                                                                          |
| 22:05 | xrpl-validator | 01aa737 | flow: slice 2 (part 2) — BookStep: forEachOffer, revImp, fwdImp, limitStepIn/Out, consumeOffer                                                                             |
| 22:05 | xrpl-validator | a904cff | flow: slice 2 (part 1) — BookTip and FlowOfferStream, TOffer, Quality::ceil_in/out                                                                                         |
| 22:05 | xrpl-validator | c195d83 | flow: slice 1b — STAmount rounding port, pinned by fuzz; slice 2 opens with view.rs                                                                                        |
| 22:05 | xrpl-validator | 1feacf6 | feat(ledger): track 2 slice 1 — flow module: amounts, QualityFunction, PaymentSandbox, the Step trait                                                                      |

| Time  | Repo           | Commit  | Subject                                                                                                                           |
|-------|----------------|---------|-----------------------------------------------------------------------------------------------------------------------------------|
| 22:15 | xrpl-validator | 9743159 | flow: slice 4 (part 1) — DirectStepI, XRPEndpointStep, rippleCredit/rippleSend/accountSend over the sandbox, XRP deferred credits |
| 22:19 | xrpl-validator | e0fa870 | flow: slice 4 (part 2) — StrandFlow (single- and multi-strand flow, ActiveStrands, limitOut) and PaySteps (toStrand / toStrands)  |
| 22:26 | xrpl-validator | 3f5a90e | flow: slice 4 (part 3) — Flow.cpp's flow(), rippleCalculate, and the Payment dispatch behind XRPL_FLOW_ENGINE=port                |